

Hackathon Project Phases Template

Project Title:

Flavour Fusion: AI-Driven Recipe Blogging

Team Name:

Fusion Fanatics

Team Members:

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Phase-1: Brainstorming & Ideation

Objective:

Develop an AI-powered recipe blogging tool that generates creative, personalized, and SEO-friendly recipes using AI while enhancing user engagement with fun and interactive features.

Key Points:

- 1. Problem Statement:**
 - o Food bloggers and content creators spend significant time crafting detailed recipes and optimizing them for SEO.
 - o Home cooks struggle to find unique, personalized, and well-structured recipes that match their taste preferences and dietary needs.
- 2. Proposed Solution:**
 - o An AI-powered application using GPT-4/Llama 2 to generate detailed and engaging recipes.
 - o Ingredient substitutions and fusion suggestions based on user preferences.
 - o Fun elements like jokes and trivia during content generation.
- 3. Target Users:**
 - o Food bloggers and content creators.

- o Home cooks and food enthusiasts.
- o Restaurants and food businesses.
- o Dietitians and nutritionists.
- o SEO and digital marketing professionals.

4. **Expected Outcome:**

- o AI-powered recipe content creation with minimal effort.
 - o Increased engagement and creativity in food blogging.
 - o Personalized and fusion recipes for diverse audiences.
 - o Improved SEO and content visibility.
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Phase-2: Requirement Analysis

Objective:

Define the technical and functional requirements for Flavour Fusion.

Key Points:

1. **Technical Requirements:**

- o **Programming Language:** Python
- o **Frontend:** Streamlit Web Framework
- o **Backend:** FastAPI
- o **Database:** PostgreSQL / Firebase Firestore
- o **AI/NLP:** GPT-4, Llama 2, or Hugging Face Transformers
- o **Hosting:** Streamlit Sharing / AWS Lambda
- o **Auth & Security:** Firebase Auth, OAuth, JWT

2. **Functional Requirements:**

- o AI-generated SEO-optimized recipes.
- o Ingredient substitutions and fusion suggestions.
- o User authentication and profile management.
- o Social media sharing options.
- o Fun engagement features like jokes/trivia.

3. **Constraints & Challenges:**

- o Ensuring AI-generated recipes are accurate and practical.

- o Avoiding AI-generated plagiarism/copyright issues.
 - o Optimizing AI-generated content for SEO and uniqueness.
 - o Handling user data securely.
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Phase-3: Project Design

Objective:

Develop the architecture and user flow of the application.

Key Points:

1. System Architecture:

- o User enters recipe-related query via UI.
- o Query is processed using AI (GPT-4/Llama 2).
- o AI model generates a unique and structured recipe.
- o Backend stores generated recipes in the database.
- o Frontend displays AI-generated recipes with user customization options.

2. User Flow:

- o Step 1: User enters a dish name and preferences.
- o Step 2: AI generates a structured recipe with variations.
- o Step 3: User reviews, customizes, and saves the recipe.
- o Step 4: User shares content or exports for blogging.

3. UI/UX Considerations:

- o Simple and intuitive Streamlit interface.
 - o Clean layout with interactive recipe adjustments.
 - o Dark mode/light mode for better readability.
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Phase-4: Project Planning (Agile Methodologies)

Objective:

Break down development tasks for efficient completion.

Sprint Planning with Priorities:

Sprint Task	Priority	Duration	Deadline	Assigned To	Dependencies	Expected Outcome
Environment Setup & AI Model Integration	🔴 High	6 hours	End of Day 1	Member 1	Python, AI API Key	AI model working
Frontend UI Development	🟡 Medium	3 hours	End of Day 1	Member 2	API response format	Basic UI with input fields
Recipe Generation & Personalization	🔴 High	4 hours	Mid-Day 2	Member 1 & 2	AI model, UI setup	AI-generated recipes with variations
Error Handling & Debugging	🔴 High	2 hours	Mid-Day 2	Member 1 & 4	API logs, UI inputs	Improved API stability
Testing & UI Enhancements	🟡 Medium	3 hours	Mid-Day 2	Member 2 & 3	API response, UI layout	Responsive UI & better experience
Final Presentation & Deployment	🟢 Low	2 hours	End of Day 2	Entire Team	Working prototype	Demo-ready project

Phase-5: Project Development

Objective:

Implement core features of Flavour Fusion.

Key Points:

1. Technology Stack Used:

- o **Frontend:** Streamlit
- o **Backend:** FastAPI
- o **Database:** PostgreSQL / Firebase Firestore
- o **AI Model:** GPT-4 / Llama 2

2. Development Process:

- o Implement AI key authentication and OpenAI API integration.
- o Develop recipe customization and fusion suggestions.
- o Optimize AI responses for creativity and accuracy.

3. Challenges & Fixes:

- o **Challenge:** Slow API response times.
Fix: Implement caching for frequently queried recipes.

- o **Challenge:** Ensuring unique recipe generation.
Fix: Fine-tune AI model responses.

Phase-6: Functional & Performance Testing

Objective:

Ensure that Flavour Fusion works as expected.

Test Case ID	Category	Test Scenario	Expected Outcome	Status	Tester
TC-001	Functional	Generate a vegan recipe	AI should return a structured recipe	✅ Passed	Tester 1
TC-002	Functional	Substitute ingredients for gluten-free users	AI should provide accurate alternatives	✅ Passed	Tester 2
TC-003	Performance	AI response time under 500ms	AI should generate content quickly	⚠ Needs Optimization	Tester 3
TC-004	Bug Fixes	Fixed incorrect recipe formatting	Data should be structured correctly	✅ Fixed	Developer
TC-005	Final Validation	Ensure UI is responsive	UI should work on mobile & desktop	❌ Failed - Needs Fixing	Tester 2
TC-006	Deployment	Host app on Streamlit Sharing	App should be accessible online	✅ Deployed	DevOps

Key Points:

- **Test Cases Executed:**
 - o AI-generated recipe accuracy
 - o Ingredient substitution reliability
 - o UI responsiveness and accessibility
 - o SEO optimization checks
- **Bug Fixes & Improvements:**
 - o Improved AI logic for recipe steps
 - o Fixed database caching issues

- **Final Validation:**
 - o Ensures AI-generated recipes are readable, creative, and structured correctly
- **Deployment:**
 - o Hosted on Streamlit Sharing
 - o AWS Lambda for AI processing
 - o Database hosted on Firebase Firestore/PostgreSQL

Final Submission

1. Project Report Based on the template.
 2. Demo Video (3-5 Minutes).
 3. GitHub/Code Repository Link.
 4. Presentation.
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